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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : CS-503

DISCRETE MATHEMATICS

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

   i) The numeric sequence of the genetic function $\frac{x^5+1}{x+1}$ is

   a) 1,–1,1,–1 b) 1 –1 +1 –1 +1 ...

   c) 1, 1, 1, 1 d) none of these.

   ii) A tree has 21 vertices, then $\chi(G)$ is

   a) 20 b) 10

   c) 40 d) none of these.

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iii) The chromatic polynomial of Peterson graph is

a) 3 b) 4

c) 5 d) none of these.

iv) The number of ways that a tree with 5 vertices can be coloured by 4 colours is

a) 324 b) 350

c) 20 d) none of these.

v) Let $L(x, y)$ be the statement "x likes y" and domain for x, y consists of all people in the world. Then we can express "Nobody likes everybody" as

a) $\forall x \exists y L(x,y)$ b) $\forall x \exists y \neg L(x,y)$

c) $\forall x \forall y \neg L(x,y)$ d) none of these.

vi) A non-empty finite poset has

a) at most one greatest element

b) at most one least element

c) either (a) or (b)

d) both (a) and (b).

vii) The solution of the recurrence relation $S_n = 2S_{n-1}$ with $S_0 = 1$ is $S_n =$

a) 2^n b) 2^{n-1}

c) 2^{n+1} d) none of these.

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viii) If p : "Anil is rich" and q : "Kajal is poor", then the symbolic form of the statement "Either Anil or Kajal is rich" is

a) $p \vee p$ b) $p \vee \neg q$

c) $\neg p \vee q$ d) $\neg (p \wedge q)$.

ix) The total number of ways in which a null graph with 5 vertices can be properly coloured with 3 colours is

a) 125 b) 243

c) 225 d) none of these.

x) The number 8955795758 is divisible by

a) 7 only b) 13 only

c) 7 or 13 or 37 d) none of these.

xi) $\neg p \vee (q \wedge \neg p)$ is equivalent ($\equiv$) to

a) q b) $\neg q$

c) p d) $\neg p$.

xii) A complemented, distributed lattice is also called a Boolean Algebra. Consider a set $S = \{a, b, c\}$ and let $M = \wp(S)$ be the power set of S. Consider the inclusion (subset) relation '$\subseteq$'. Then $(M, \subseteq)$ is

a) not a partial ordering

b) a partial ordering but not a lattice

c) a lattice but not a Boolean algebra

d) a Boolean algebra.

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GROUP - B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. C_9 is a cycle (*i.e.*, a circular chain) with the nine vertices $a, b, c, d, e, f, g, h, i$. How many distinct maximal matchings of size four in C_9 contain the edge ab? http://www.makaut.com

3. Solve the difference equation $y_{n+4} - y_n = 2^n$.

4. Consider $\mathbb{N}$, the set of all natural numbers and a relation '$\leq$' defined on $\mathbb{N}$ by $a \leq b$ holds if and only if $a \mid b$, $\forall a, b \in \mathbb{N}$. Then show that $(\mathbb{N}, \leq)$ is a poset.

5. In an area the custom is that girls give their choice to whom they will marry. One such choice is given below. Give your decision for matching and decide whether each girl will be married within their choice.

   $g_1 \rightarrow (b_1, b_3, b_5), g_2 \rightarrow (b_1, b_2, b_4), g_3 \rightarrow (b_3, b_4, b_5),$

   $g_4 \rightarrow (b_2, b_3, b_5), g_5 \rightarrow (b_2, b_3, b_4)$.

6. Find whether the argument $p \rightarrow \sim q, r \rightarrow q, r \mid = \sim p$ is valid or not.

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GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. a) If $S = \{ 1, 2, 3 \}$ then draw the Hasse diagram of the poset $(\wp(S), \subseteq)$, where $\wp(S)$ is the power set of S. 5

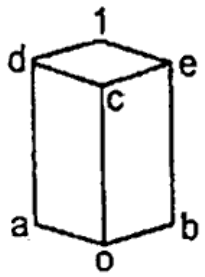
   b) Using the laws of propositional logic show that $\models (p \wedge q) \rightarrow (p \vee q)$. 5

   c) Using generating function solve the difference equation $y_{n+2} - 5y_{n+1} + 6y_n = 0$ when $y_0 = 1$, $y_1 = 0$ 5

8. a) Show that every bipartite graph is 2 chromatic. 5

   b) Find integers u and v satisfying
   $gcd\,(272, 119) = 272u + 119v$. 5

   c) Show that the poset given in the following Hasse diagram is a lattice. Is it distributive and complemented ? 5



9. a) Write down the quantifiers in predicate calculus and symbolize the following statements :

   i) Every rational number is real number.

   ii) There exists a number which is prime. 2 + 3

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b) Symbolize the following :
If either George enrols or Harry enrols then Ira does not enrol. Either Ira enrols or Harry enrols. If either Harry enrolls or George does not enrol then Jim enrols. George enrols. Therefore either Jim enrols or Harry does not enrol. 5

c) What is a formula in proportional calculus ? What is a tautology ? State and prove Duality theorem. 2 + 1 + 2

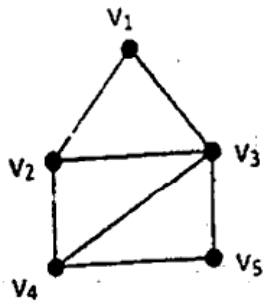
10. a) Find the non-negative integral solution of the equation $x + y + z = 18$. 5

b) State Generalized Piegonhole principle. Find the minimum number of teachers in an institute to be sure that four of them are born in the same month. 5

c) Find the total number of integers lying between 1 and 1000 that are divisible by at least one of 2, 3, 7. 5

11. a) A connected planar graph with n vertices and e edges has $e - n + 2$ regions. 5

b) Find the chromatic polynomial of the graph given below : 5



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c) Show that the Chromatic number of a complete graph with n vertices (K_n) is n. 5

12. a) Prove that every tree has at most one perfect matching. 5

b) Give a direct proof that if x and y are numbers such that $5x+15y=116$, then either x and y is not an integer. 5

c) Show that the Peterson graph does not contain two disjoint perfect matching. 5

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